Example 8.30

Find the probability of drawing a queen, a king and a knave (Jack) in that order from a pack of cards in three consecutive draws, the card drawn not being replaced.

Solution:

Let A : the card drawn is a queen

B: the card drawn is a king

C: the card is drawn is a knave(jack)

P (drawing a queen card) = $P(A) = \frac{4}{52}$

P (drawing a king card given that a queen card has been already drawn)

$$= \text{P(B/A)} = \frac{4}{51}$$

P (drawing a knave card given that a queen and a king cards have been drawn)

$$= \text{P(C/AB)} = \frac{4}{50}$$

Since they are dependent events, the required probability of the compound event is

$$\text{P(ABC)} = \text{P(A) P(B/A) P(C/AB)}$$

$$= \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} = \frac{64}{132600}$$

$$= 0.00048$$

8.3.4 Baye's Theorem

If $E_1, E_2, E_3 ..., E_n$ are a set of n mutually exclusive and collectively exhaustive events with $P(E_i) \neq 0$ (i = 1,2,3…, n), then for any arbitrary event A which is associated with sample space $S = \bigcup_{i=1}^{n} E_i$ such that $P(A) > 0$, we have

$$P(E_i / A) = \frac{P(E_i)P(A / E_i)}{\sum_{i=1}^{n} P(E_i)P(A / E_i)}; i = 1,2,3,\ldots, n$$

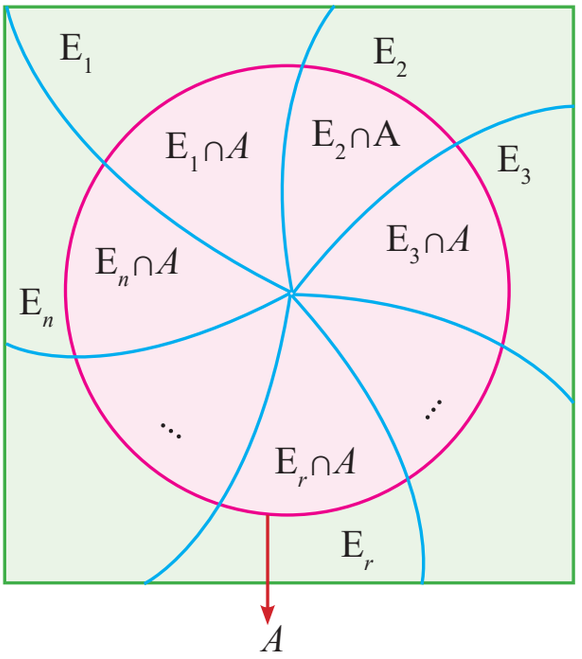


Fig. 8.2

where $P(A) = \sum_{i=1}^{n} P(E_i)P(A / E_i)$

Example 8.31

Bag I contains 3 red and 4 blue balls while another Bag II contains 5 red and 6 blue balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from second Bag.

Solution:

Let E_1 be the event of choosing the first bag, E_2 the event of choosing the second bag and A be the events of drawing a red ball. Then $P(E_1) = P(E_2) = \frac{1}{2}$.

Also $P(A/E_1) = P$ (drawing a red ball from Bag I) = $\frac{3}{7}$

and $P(A/E_2) = P$ (drawing a red ball from Bag II) = $\frac{5}{11}$.

Now, the probability of drawing a ball from Bag II, being given that it is red, is $P(E_2/A)$.

By using Baye's theorem, we have

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